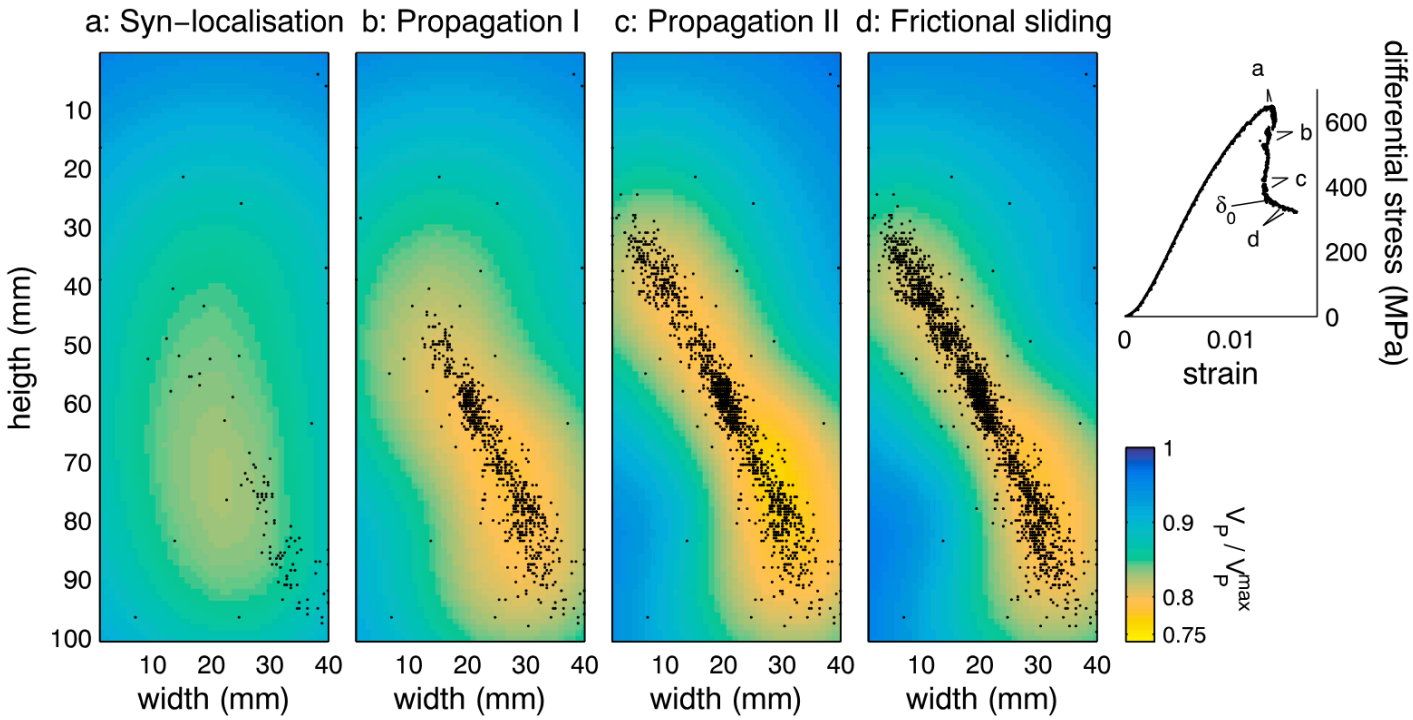
Can we hear a rock approaching failure?

From microcrack initiation to AE detection during a UCS experiment

One experiment • three levels of observation: stress–strain, cracking, acoustic signals

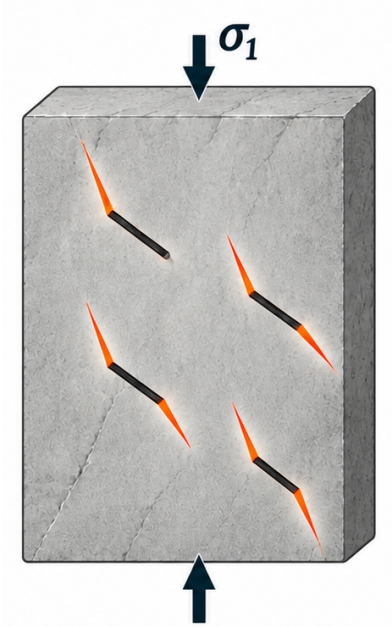


How do many small cracks interact?

An isolated-crack model is only the start of a many-crack failure process.

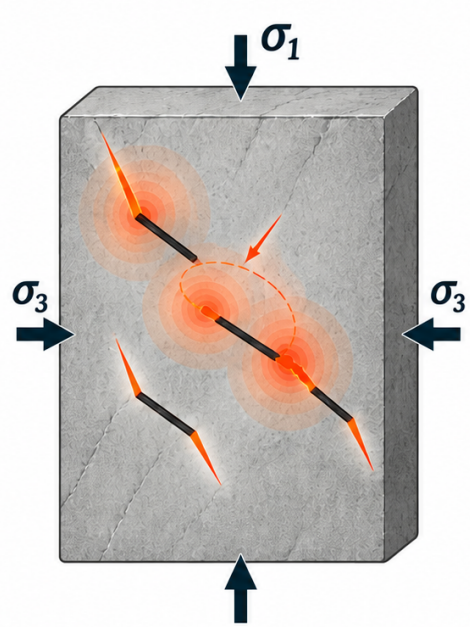
I · Isolated

Stable growth near favorably oriented flaws



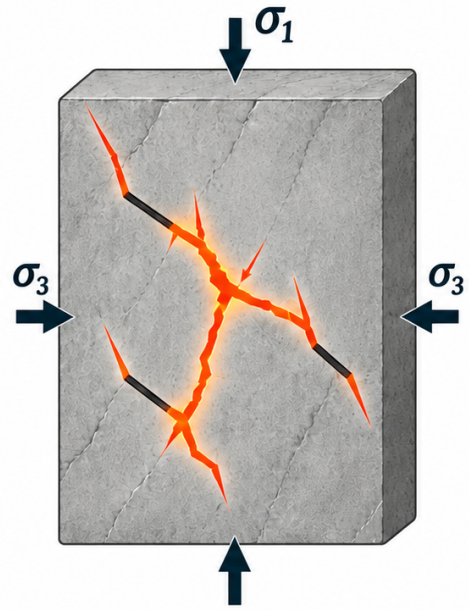
II · Interaction

Crack-tip stress fields overlap



III · Coalescence

Connected fracture paths develop

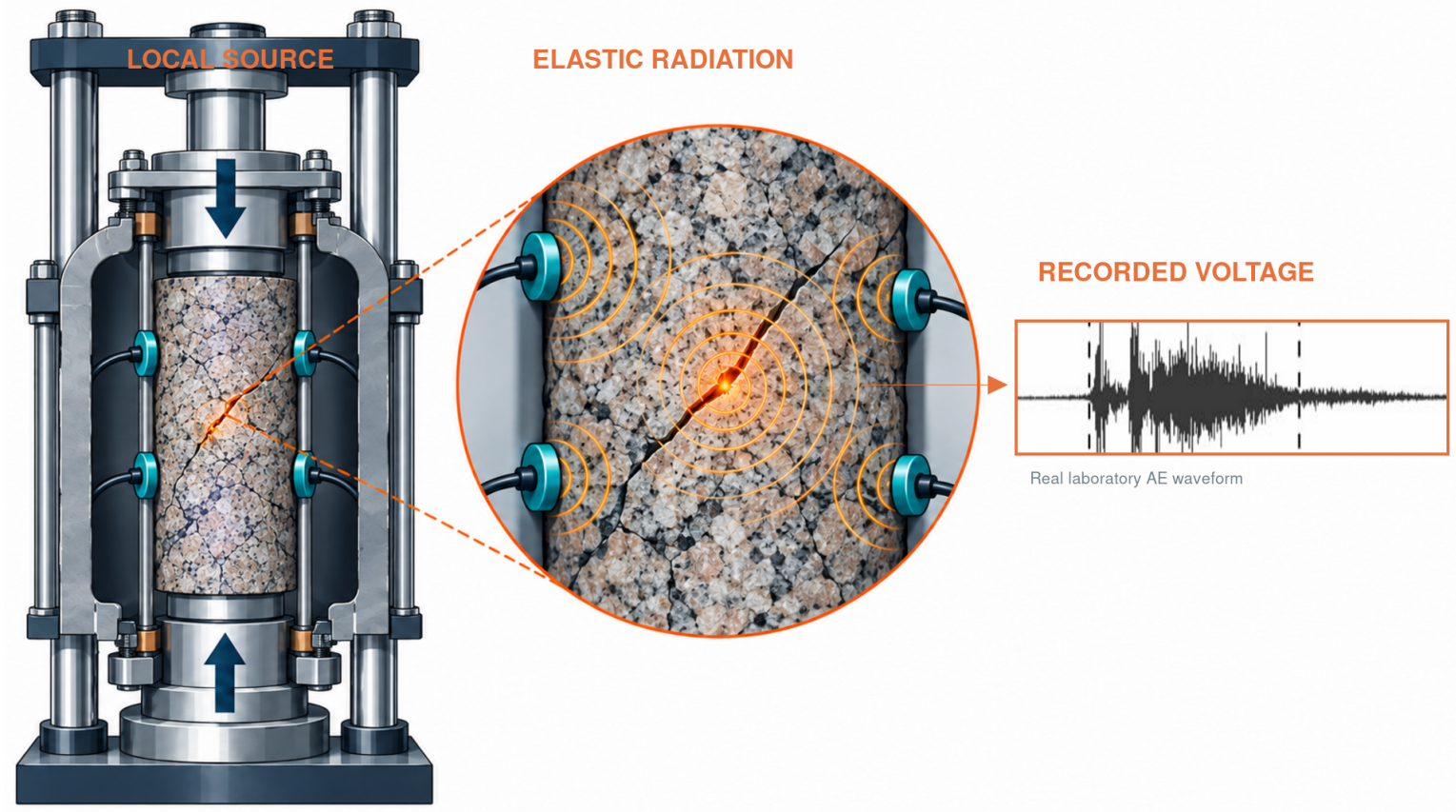


Conceptual progression. Wong & Einstein (2009); Moore & Lockner (1995); McBeck et al. (2026).

What should you be able to do after this lecture?

1. Explain why cracks develop under compression
2. Distinguish crack nucleation, propagation, coalescence and localization
3. Explain what generates and what limits an AE measurement
4. Choose a defensible detection threshold and compute AE rate
5. Predict the main AE observations during our UCS laboratory

A microfracture is a source of transient elastic waves



1 Local source

crack opening / sliding / grain breakage

2 Elastic radiation

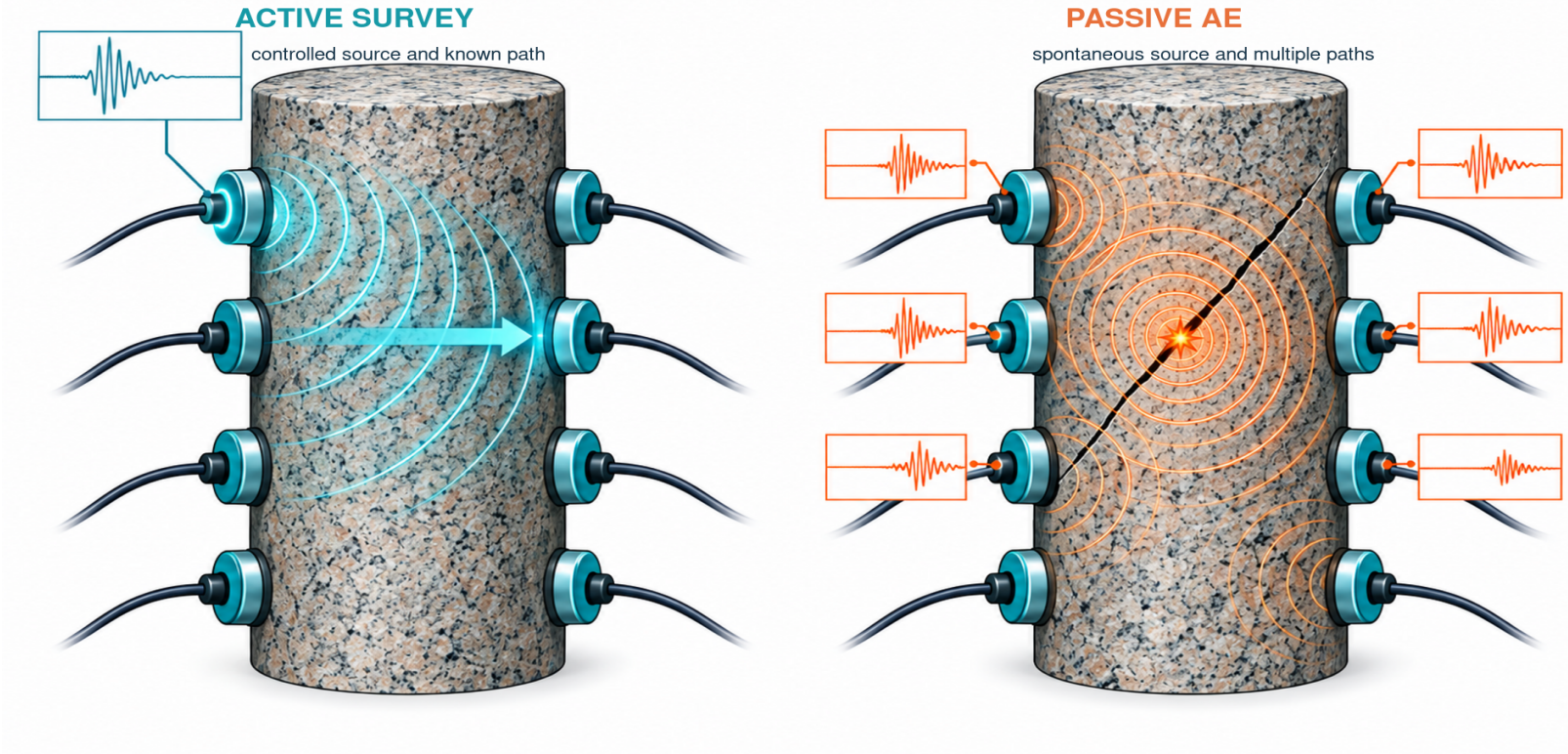
transient P- and S-wave energy leaves the source

3 Recorded voltage

a sensor converts surface motion into an electrical waveform

AE sensors record voltage generated by arriving elastic waves—not cracks directly.

Active ultrasonics versus passive AE — only the essential difference



Why use both?

Active measurement: controlled source; probes the evolving material along a path.

Passive AE: unknown source time/location; detects only events that radiate sufficiently strong waves.

One tells us how the rock transmits waves; the other detects rapid local sources.